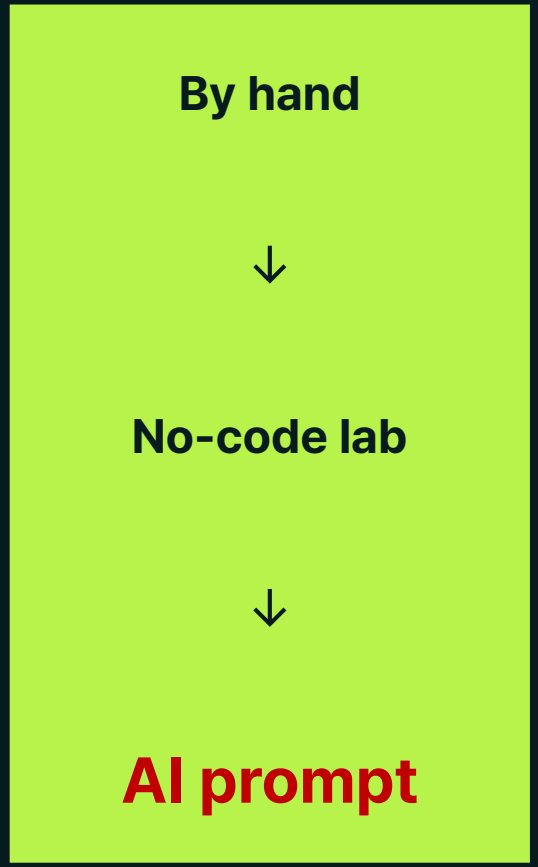
Causality · XAI · Fairness Workbook

Step-by-step solutions you can follow with a calculator



Financial ML · Week 14 Workbook

The 1.2-pt mention gap comes from the mix of up days

Up day 2%, quiet day 0% — up days: 4 of 5 high, 1 of 5 low

Step 1 · Group means

$$\text{high-mention days} = \frac{4(2\%) + 1(0\%)}{5} = 1.6\%, \qquad \text{low-mention days} = \frac{1(2\%) + 4(0\%)}{5} = 0.4\%$$

Step 2 · SHAP (mean over orders)

$$\phi_{\text{debt}} = \frac{(0.70 - 0.40) + (0.65 - 0.30)}{2} = 0.325$$

$$\phi_{\text{income}} = \frac{(0.30 - 0.40) + (0.65 - 0.70)}{2} = -0.075$$

$$\text{baseline} + \phi_{\text{debt}} + \phi_{\text{income}} = 0.40 + 0.325 - 0.075 = 0.65$$

SHAP 0.325 splits the model's output, not a causal effect

The denominator changes how the group gap looks

Groups of 100, 80 qualified, no unqualified approved

Step 1 · Metrics by group

$$\text{overall approval rate}: \quad \frac{60}{100} = 60\%, \qquad \frac{40}{100} = 40\%$$

$$\text{TPR (qualified approval)}: \quad \frac{60}{80} = 75\%, \qquad \frac{40}{80} = 50\%$$

$$\text{accuracy}: \quad \frac{60+20}{100} = 80\%, \qquad \frac{40+20}{100} = 60\%$$

Approval rate divides by all; TPR by the qualified

Dropping ZIP codes need not remove the group gap

Separate prediction, explanation, causation in numbers

Notepad and calculator

1. Where

Use a notepad and calculator.

2. Input

Write both group sizes.

3. Output

Tabulate approval, TPR and accuracy.

4. Check

Say who each denominator covers.

Prompt for the AI chat

Define prediction, model explanation and causal effect in one sentence each. Show with numbers how market regime drives both social mentions and returns. Interpret SHAP 0.40+0.325−0.075=0.65 and why it is not causal. Finally, compute approval rate and TPR for groups A and B with their own denominators.

CALCULATE BY HAND.

CHECK THE DENOMINATOR.

THEN TRUST THE NUMBER.

Calculate once by hand
and the assumptions show.